Q1 correct
Q2 -25
Q3 -10
Q4 -20

Homework 7 - Deadline December 20th

Exercise 1. a) ***10 points.*** Solve $\int \frac{1}{(x^2+4)^2} dx$.

1. b) ***15 points.*** Compute

$$\int \frac{x^4+8x^2+x+16}{x(x^2+4)^2} dx.$$

a) $\int \frac{1}{(x^2+4)^2} dx$

Assume $x = 2\tan\theta$, $dx = 2\sec^2\theta \, d\theta$

$$\int \frac{1}{(x^2+4)^2} dx = \int \frac{1}{(4\tan^2\theta+4)^2} dx$$

$$= \int \frac{2\sec^2\theta}{16\sec^4\theta} d\theta$$

$$= \int \frac{1}{8\sec^2\theta} d\theta$$

$$= \frac{1}{8}\int \frac{1}{\sec^2\theta} d\theta$$

$$= \frac{1}{8}\int \cos^2\theta \, d\theta$$

$$= \frac{1}{16}\int 1+\cos 2\theta \, d\theta$$

$$= \frac{1}{16}\left(\theta + \frac{1}{2}\sin 2\theta\right) + C$$

$$\int \frac{1}{(x^2+4)^2} dx = \frac{1}{16}\left(\arctan\frac{x}{2} + \frac{1}{2}\sin\left(2\arctan\left(\frac{x}{2}\right)\right)\right] + C$$

b) $\int \frac{x^4+8x^2+x+16}{x(x^2+4)^2} dx$

$$= \int \frac{x^4+8x^2+16+x}{x(x^4+16+8x^2)} dx$$

$$= \int \frac{1}{x} dx + \int \frac{1}{(x^2+4)^2} dx$$

$$= \ln|x| + \frac{1}{16}\left(\arctan\frac{x}{2} + \frac{1}{2}\sin\left(2\arctan\left(\frac{x}{2}\right)\right)\right] + C$$

Exercise 2. ***25 points.*** Find the volume of the solid enclosed by the revolution surface obtained by the rotation of the curve $x = e^y$ around the x-axis on $1 \leq y \leq 3$.

We know $V = \pi \int_a^b [f(y)]^2 \, dy$

Since $f(y) = e^y$, $V = \pi \int_1^3 e^{2y} \, dy$

$= \pi \left(\frac{1}{2} e^{2\times 3} - \frac{1}{2} e^{2\times 1} \right)$

$= \frac{1}{2}\pi (e^6 - e^2)$

2.1

Exercise 3. ***25 points.*** Find the volume of the solid enclosed by the revolution surface obtained by the rotation of the curve $y = \cos(x^2)$ around the y-axis on $\frac{\sqrt{\pi}}{2} \le x \le \sqrt{\frac{\pi}{2}}$.

We know $V = \int_a^b 2\pi x f(x)\,dx$

$$= \int_{\frac{\sqrt{\pi}}{2}}^{\sqrt{\frac{\pi}{2}}} 2\pi x \cos(x^2)\,dx$$

$$= 2\pi \int_{\frac{\sqrt{\pi}}{2}}^{\sqrt{\frac{\pi}{2}}} x \cos(x^2)\,dx$$

Assume $u = x^2$, $du = 2x\,dx$

when $x = \frac{\sqrt{\pi}}{2}$, $u = \frac{\pi}{4}$,

$x = \sqrt{\frac{\pi}{2}}$, $u = \frac{\pi}{2}$

$$\int_{\frac{\sqrt{\pi}}{2}}^{\sqrt{\frac{\pi}{2}}} x\cos(x^2)\,dx = \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos u \cdot \frac{1}{2}\,du$$

$$= \frac{1}{2}\int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \cos u\,du$$

$$= \frac{1}{2} \sin u \Big|_{\frac{\pi}{4}}^{\frac{\pi}{2}}$$

$$= \frac{1}{2}\left(1 - \frac{\sqrt{2}}{2}\right)$$

In conclusion, $V = \pi\left(1 - \frac{\sqrt{2}}{2}\right)$

Exercise 4. a) ***10 points.*** Solve

$$\int \frac{\left(2\tan^2(x)+1\right)\sec^2(x)}{(\sec^2(x)-1)(\tan^2(x)+1)}dx$$

b) ***15 points*** Use 1st a substitution and then integration by parts, to solve

$$\int \frac{\sin\left(\ln\sqrt{1+x^{-4/5}}\right)}{(1+x^{-4/5})(x+x^{9/5})}dx$$

a) We know $\sec^2(x)-1 = \tan^2(x)$

$$x(1+x^{-\frac{4}{5}})(1+x^{\frac{4}{5}})$$

$$\int \frac{(2\tan^2(x)+1)\sec^4(x)}{(\sec^2(x)-1)(\tan^2(x)+1)} = \int \frac{(2\tan^2(x)+1)\sec^2(x)}{\tan^2(x)(\tan^2(x)+1)}dx$$

$$= \int \frac{(2\tan^2(x)+1)(1+\tan^2(x))}{\tan^2(x)(\tan^2(x)+1)}dx$$

$$= \int \frac{2\tan^2(x)+1}{\tan^2(x)}dx$$

$$= \int 2\,dx + \int \frac{1}{\tan^2(x)}dx$$

$$= 2x + \frac{1}{2}\sec^2(x) + C$$

Index of comments

2.1 Q2
x=e^y, so ln(x)=y
e≤x≤e^3,
Volume= integration of (π*((ln(x))^2))dx from x=e to x=e^3
V=π[5(e^3)-e]

3.1 sqrt(2)/8*pi^2-pi/sqrt(2)+pi

"You can look at the cross-section of xz plane, which are circles. And integrate along the y axis"